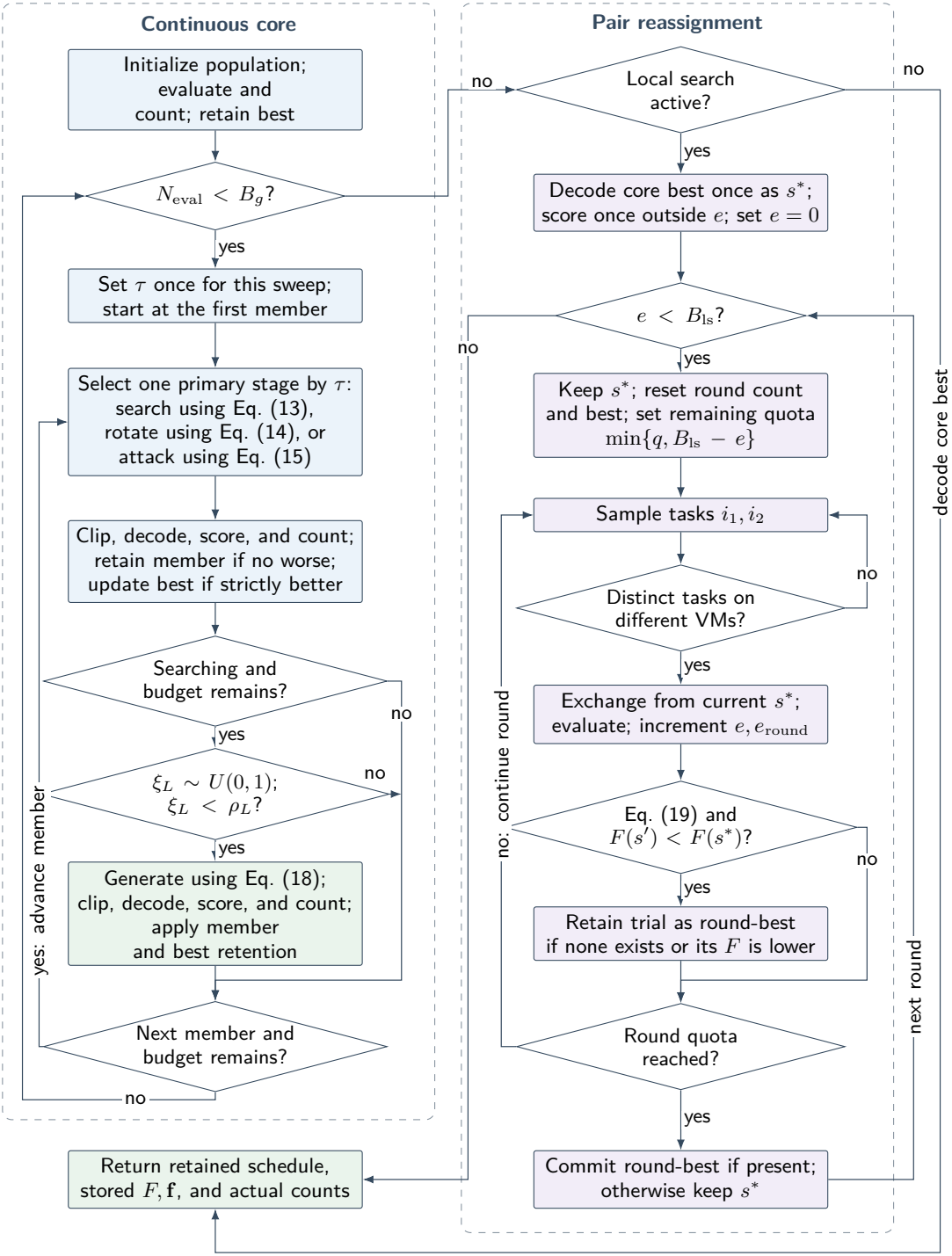

Continuous core
Initialize population; evaluate and count; retain best
$N_{\text{eval}} < B_g$?
yes
Set τ once for this sweep; start at the first member
Select one primary stage by τ: search using Eq. (13), rotate using Eq. (14), or attack using Eq. (15)
Clip, decode, score, and count; retain member if no worse; update best if strictly better
Searching and budget remains?
yes
no
$\xi_L \sim U(0,1)$; $\xi_L < \rho_L$?
yes
no
Generate using Eq. (18); clip, decode, score, and count; apply member and best retention
Next member and budget remains?
yes: advance member
no
Return retained schedule, stored $F, \mathbf{f}$, and actual counts
Pair reassignment
Local search active?
no
yes
Decode core best once as s^*; score once outside e; set $e = 0$
$e < B_{\text{ls}}$?
no
yes
Keep s^*; reset round count and best; set remaining quota $\min\{q, B_{\text{ls}} - e\}$
Sample tasks i_1, i_2
Distinct tasks on different VMs?
no
yes
Exchange from current s^*; evaluate; increment e, e_{round}
Eq. (19) and $F(s') < F(s^*)$?
yes
no
Retain trial as round-best if none exists or its F is lower
Round quota reached?
no: continue round
yes
Commit round-best if present; otherwise keep s^*
next round
decode core best